

Carbon Continuity After Wildfire

A dynamic portfolio framework and regenerative coupling threshold for natural carbon assets

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Abstract

Wildfire challenges forest-carbon accounting because a disturbance changes several carbon pools at once: it emits carbon, transfers living biomass into dead and pyrogenic pools, alters future uptake, and can impair soil and hydrologic conditions that govern regeneration. A stock-only permanence test therefore omits both durable post-fire carbon and foregone future uptake. We propose a Carbon Continuity Framework that keeps physical carbon accounting separate from ecosystem co-benefits while evaluating the persistence of carbon service across disturbance-recovery cycles. Its formal core is a two-pool local model with living carbon L , durable carbon D , within-pool retention $a, d \in [0, 1)$, living-to-durable conversion $p > 0$, and durable-support-mediated regeneration $r > 0$. A strictly positive portfolio can be nondecreasing through one standardized cycle if and only if

$$rp \geq (1 - a)(1 - d),$$

or equivalently the Carbon Continuity Number $\mathcal{R}_C = rp/[(1 - a)(1 - d)]$ is at least one. The result identifies a coupling threshold rather than an optimal treatment: coefficients must be estimated by site, biome, severity, and time horizon. An exact rational-number gate passed 20 of 20 checks, including 25,000 sufficiency cases, 25,000 adversarial necessity cases, boundary and negative controls, and a capped illustrative simulation. The six frozen Lean 4 theorems were completed by Aristotle and then independently rebuilt and audited locally with zero proof holes and only the permitted foundational axioms. We translate the model into measurable post-fire pools, a finite-horizon Carbon Continuity Index, an intervention hierarchy, and preregisterable field tests. The framework does not claim that fire is carbon-neutral, that pyrogenic carbon offsets combustion, or that predicted resilience is a carbon credit. It offers a falsifiable bridge between carbon-pool dynamics, regeneration, and disturbance-risk accounting.

Keywords: wildfire; forest carbon; permanence; resilience; pyrogenic carbon; regeneration; dynamic accounting; carbon markets; formal verification

1 Introduction

Terrestrial carbon is not geologically permanent. Forest carbon can be lost or redistributed by wildfire, drought, insects, windthrow, harvest, and land-use change, and these risks change with climate. The IPCC therefore distinguishes the durability of biological storage from longer-lived geological reservoirs [3]. Recent work finds that major forest-carbon protocols materially underestimate climate-driven disturbance risk [1]; earlier synthesis likewise identified climate risk as a central constraint on forests' mitigation potential [2]. An actuarial analysis of California's compliance program concluded that observed wildfire losses had consumed at least 95% of the buffer contribution intended to cover 100 years of fire risk by the end of 2021 [4]. These findings do not imply that forests lack climate value. They imply that risk, recovery, and time cannot remain external to the accounting model.

The opposite simplification is also unsafe. Fire is not only an instantaneous emission. It transfers carbon among living biomass, standing dead material, down wood, litter, mineral soil, pyrogenic carbon, harvested products, and the atmosphere. A global synthesis estimated substantial pyrogenic-carbon production by landscape fire [5], yet later modeling showed that the sign of the full fire carbon balance remains unresolved without better constraints on pyrogenic-carbon mineralization and legacy fluxes [6]. Post-fire uptake matters as well: California ecosystems

in a century-scale fire chronosequence recovered gross primary productivity on average after roughly 12 years, but recovery was slower following more recent, larger, and more severe fires [7]. Frequent fire can also reduce soil carbon and nitrogen and depress productivity over decades [8].

This paper asks a narrow question: *what minimum mathematical structure is needed to distinguish a carbon stock from a carbon asset that can continue to deliver carbon service through recurrent disturbance?* We make four contributions.

1. We define a carbon-pool accounting boundary that records emissions, transfers, retained carbon, and future uptake without mixing tonnes of carbon with water or biodiversity units.
2. We derive an exact two-pool coupling threshold for local carbon continuity and state its limitations.
3. We propose a finite-horizon Carbon Continuity Index (CCI) and an empirical parameterization protocol suitable for preregistration.
4. We translate the framework into cautious management and carbon-market design principles, explicitly separating a risk metric from a credit unit.

The nearest literature already includes risk-sensitive carbon accounting, landscape simulation, resilience-oriented management, and diversified natural- climate-solution portfolios [11, 12, 1]. We do not claim priority for those ideas. The bounded novelty claim is the explicit coupled-pool threshold and its use as a falsifiable bridge between post-fire partitioning and regenerative capacity.

2 Carbon continuity as an accounting boundary

2.1 A transfer-plus-loss event

Let C_t be the total physical carbon inside a declared project boundary at a declared time. It is the sum of mutually exclusive pools, for example

$$C_t = C_t^{\text{live}} + C_t^{\text{dead}} + C_t^{\text{litter}} + C_t^{\text{soil}} + C_t^{\text{PyC}} + C_t^{\text{HWP}}. \quad (1)$$

Atmospheric emissions and exported carbon are flows outside that boundary. Equation (1) is deliberately ordinary: continuity requires better time dynamics, not relaxed mass balance. A transfer from a killed tree to standing dead wood is not a second tonne. A harvested-wood-product pool is included only with explicit geographic, life-cycle, and decay boundaries. Co-benefits such as water retention, habitat, safety, and local revenue are reported in their own units and may enter a decision objective, but they are not added to physical carbon.

We therefore replace the draft slogan “a disturbance does not destroy a carbon asset” with a testable statement:

A disturbance is a vector-valued transfer-and-loss event. Carbon continuity depends on both the retained post-disturbance portfolio and the system’s future capacity to rebuild carbon service.

This formulation allows complete asset failure. High-severity fire can emit large stocks, kill seed sources, reduce uptake, destabilize soils, or induce a state change from forest to non-forest. Continuity is an outcome to measure, not a presumption.

2.2 The management objective

For a stewardship policy π , a finite planning horizon H , discount factor $\beta \in (0, 1]$, carbon price or shadow value $P_C(t)$, co-benefit vector E_t , and intervention cost K_t , a decision objective may be written

$$J_H(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^H \beta^t (P_C(t)C_t + v^\top E_t - K_t) \right]. \quad (2)$$

This is an economic decision functional, not a carbon-credit quantity. Physical carbon, co-benefits, and cost remain separately auditable. The expectation is over fire occurrence and severity, climate, regeneration, management response, and parameter uncertainty. A policy that maximizes standing biomass today need not maximize Eq. (2), but neither does resilience language prove that an intervention is beneficial. Each comparison needs a counterfactual baseline and uncertainty analysis.

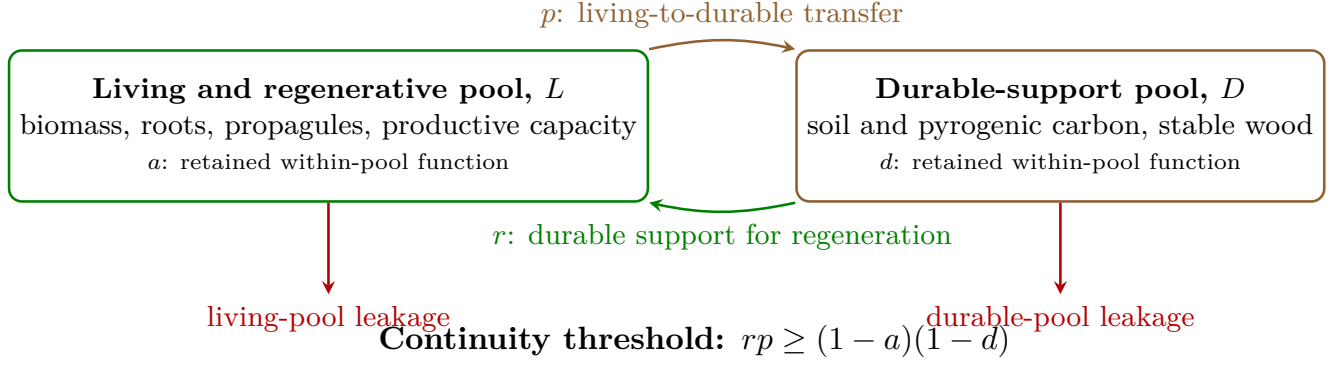


Figure 1: Two-pool model. The diagram separates physical carbon pools from drivers of the coefficients. Atmospheric and off-boundary losses are the complements of retained and transferred carbon under an explicit mass-balance protocol.

3 Two-pool regenerative coupling model

3.1 State and coefficients

For the formal core, aggregate the carbon service into two standardized, nonnegative state variables at a common cycle boundary:

- L_t : living and regenerative carbon service, including live biomass, roots, and productive capacity represented in carbon units;
- D_t : durable-support carbon, including measured soil organic carbon, pyrogenic carbon, and other explicitly bounded persistent carbon pools.

Hydrologic function, seed availability, species composition, and climate are not tonnes of carbon. They influence coefficients, especially r , and are measured as covariates. For one standardized disturbance-recovery cycle, define the local linear map

$$\begin{pmatrix} L_{t+1} \\ D_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} a & r \\ p & d \end{pmatrix}}_{M_\pi} \begin{pmatrix} L_t \\ D_t \end{pmatrix}, \quad 0 \leq a, d < 1, \quad p, r > 0. \quad (3)$$

Here a is the fraction of living carbon service retained or restored within the living channel; d is durable-pool retention; p is net conversion from living to durable carbon; and rD_t represents atmospheric uptake and living-pool recovery enabled by the durable-support state. Thus r is not a claim that soil carbon is literally converted into live biomass. The map is a local cycle-scale approximation. Carrying capacity, density dependence, changing climate, multiple severity classes, exogenous recruitment, and stochastic fire timing are suppressed until Section 5.

3.2 Carbon Continuity Theorem

Call a positive portfolio *cycle-nondecreasing* when $L > 0$, $D > 0$, and $M_\pi(L, D)^\top \geq (L, D)^\top$ componentwise.

Theorem 1 (regenerative coupling threshold). Under Eq. (3), a strictly positive cycle-nondecreasing portfolio exists if and only if

$$rp \geq (1 - a)(1 - d). \quad (4)$$

Derivation. If a positive portfolio is nondecreasing, then

$$(1 - a)L \leq rD, \quad (1 - d)D \leq pL. \quad (5)$$

Multiplying the inequalities and cancelling $LD > 0$ gives Eq. (4). Conversely, choose the explicit positive witness $L = r$ and $D = 1 - a$. The living component is held exactly:

$$ar + r(1 - a) = r. \quad (6)$$

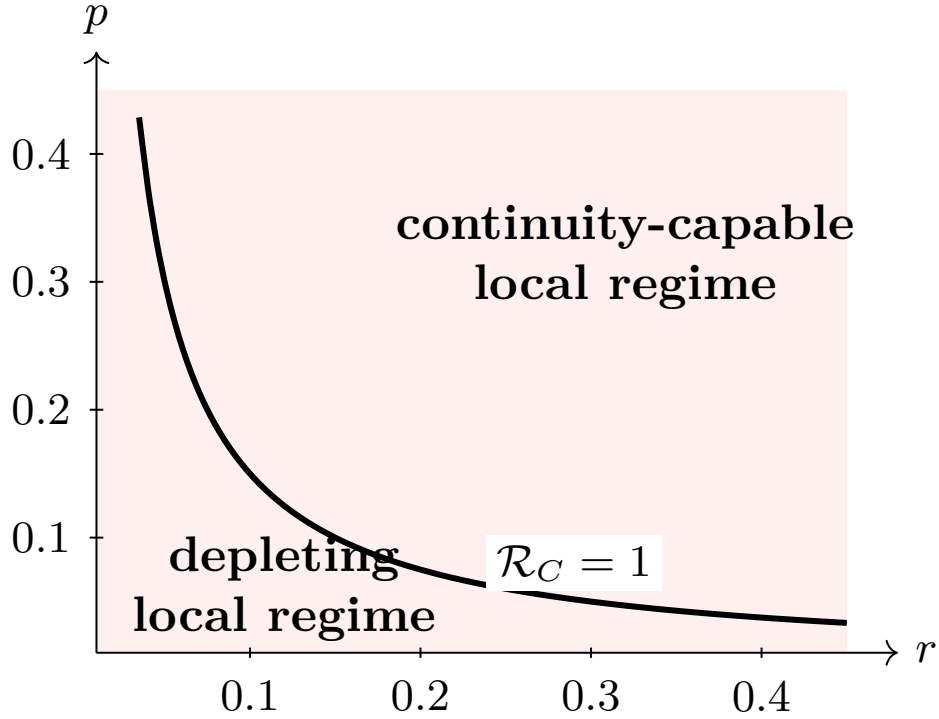


Figure 2: Illustrative phase boundary for fixed within-pool retention. This is not an empirical fire-risk map. Each coefficient must be estimated for the site, cycle definition, severity distribution, and policy.

The durable component is nondecreasing exactly when $pr + d(1 - a) \geq 1 - a$, which is Eq. (4). At equality the witness is stationary; under strict inequality its durable component grows while its living component is held.

Define the dimensionless *Carbon Continuity Number*

$$\mathcal{R}_C = \frac{rp}{(1 - a)(1 - d)}. \quad (7)$$

Then $\mathcal{R}_C = 1$ is the local boundary. For a nonnegative 2×2 matrix with $a, d < 1$, the same condition is equivalent to the dominant eigenvalue of M_π being at least one. The coupling form is more interpretable: the regenerative loop gain rp must offset the product of the two leakage terms.

3.3 What the theorem does and does not establish

The theorem establishes an exact viability condition inside a deliberately small model. It does not show that a forest grows without limit, that management can make $\mathcal{R}_C \geq 1$, that a particular biochar or erosion treatment raises either coupling, or that a cycle-nondecreasing state is globally stable. When $\mathcal{R}_C > 1$, the uncapped linear map can grow without bound; that is a signal to restore carrying capacity and nonlinear feedbacks, not a biological prediction. When $\mathcal{R}_C < 1$, the model says that no positive state can keep both aggregated pools nondecreasing without exogenous input under the chosen cycle and coefficients. It does not imply ecosystem collapse on the next calendar year.

The result is useful because it separates four levers that stock accounting can conflate. A policy can retain living function (a), retain durable carbon (d), increase net durable conversion (p), or improve the support that durable site condition provides to regeneration (r). Any claimed treatment effect must identify its lever, direction, lag, uncertainty, and possible trade-offs.

Stylized capped trajectories (illustrative, not calibrated)
normalized state

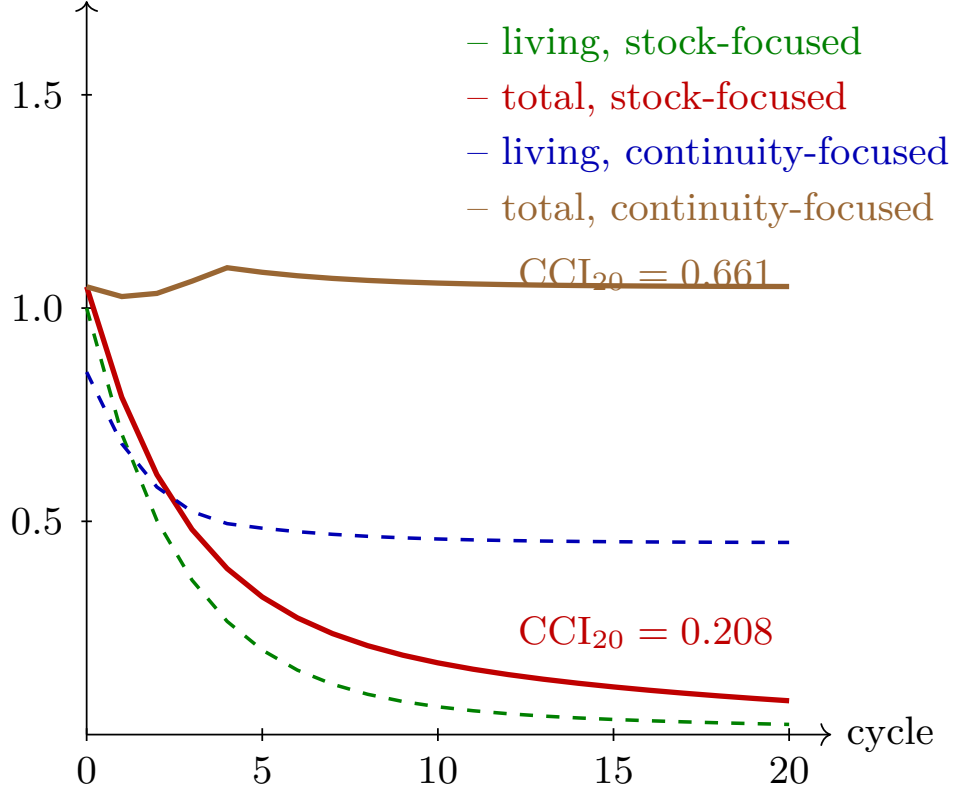


Figure 3: Illustrative capped model trajectories. The comparison is a computational example, not treatment-effect evidence.

4 A finite-horizon Carbon Continuity Index

The threshold is a local diagnostic, not a ranking metric. For a bounded or empirically simulated trajectory with declared service capacity $K_L + K_D > 0$, define

$$CCI_H(\pi) = \frac{\mathbb{E}_\pi \left[\sum_{t=0}^H \beta^t (L_t + D_t) \right]}{\sum_{t=0}^H \beta^t (K_L + K_D)}. \quad (8)$$

If states are bounded by the declared capacities, CCI lies in $[0, 1]$. A reference-relative version may divide by an unmanaged counterfactual rather than capacity; that ratio may exceed one and must not be called a fraction. Neither version is a tonne of CO_2 equivalent. Crediting still requires physical mass balance, additionality, leakage, uncertainty deduction, and a defensible permanence method.

Figure 3 illustrates why instantaneous living carbon and continuity can rank policies differently. Both stylized policies begin with the same total state, but the stock-focused case starts with more living carbon. The continuity-focused coefficients cross the coupling threshold and produce a higher 20-cycle score in a capped deterministic simulation. The numbers are chosen to exercise the model; they are not calibrated to a forest or treatment.

5 Measurement and falsification protocol

5.1 Pool measurement

At minimum, a field implementation should measure or model the mutually exclusive pools in Eq. (1) at pre-fire, immediate post-fire, and repeated recovery times. Standing dead and downed wood must not be treated as instant atmospheric emissions, but their expected decay and reburn exposure must not be ignored. Pyrogenic carbon must be measured with a declared analytical method and uncertainty; wildfire charcoal cannot be assumed to have the same properties or persistence as engineered biochar [9, 10]. National inventory methods already provide a starting vocabulary for forest, product, wildfire, and prescribed-fire accounting [17].

The cycle coefficients can be estimated from paired or hierarchical longitudinal data. For parcel i and cycle t , fit

$$\begin{pmatrix} L_{i,t+1} \\ D_{i,t+1} \end{pmatrix} = M_{i,t}(\pi, s, c, z) \begin{pmatrix} L_{i,t} \\ D_{i,t} \end{pmatrix} + u_{i,t} + \varepsilon_{i,t}, \quad (9)$$

where s is severity, c is climate, z contains site covariates, u is a declared exogenous input, and ε is residual error. Uncertainty in M must be propagated into $\Pr(\mathcal{R}_C \geq 1)$ and CCI, rather than replacing coefficients by favorable point estimates.

5.2 Preregistered predictions and stop rules

1. **Threshold prediction.** After out-of-sample parameterization, sites whose lower uncertainty bound for $\mathcal{R}_C$ exceeds one should be more likely to maintain both aggregated pools over the next standardized cycle than sites whose upper bound is below one. Refute the classifier if discrimination is no better than a preregistered standing-stock baseline.
2. **Forecast prediction.** A model using live and durable pools plus coupling should predict 5–15 year post-fire carbon service better than live biomass alone. Refute the added-value claim if blocked cross-validation does not improve a preregistered proper score after complexity penalty.
3. **Pyrogenic-pool prediction.** Explicit PyC measurement should reduce residual bias in persistent-carbon forecasts where PyC is material. Refute if the effect disappears under method uncertainty or is not portable across held-out sites.
4. **Management prediction.** A treatment may improve CCI even when it causes an immediate carbon cost, but only if the full counterfactual trajectory and uncertainty support the gain. Stop the claim when the lower confidence bound on incremental CCI is nonpositive or when biodiversity, safety, air- quality, or water constraints fail.

These predictions require real data. The present manuscript supplies no empirical estimate of a, d, p, r , no treatment effect, and no causal claim.

6 Post-fire stewardship hierarchy

Post-fire action should follow hazard and ecological diagnosis, not a universal recipe. Fire severity, soil burn severity, slope, rainfall regime, vegetation type, seed sources, invasive-species risk, public safety, and future fire behavior can reverse a generic ranking.

1. **Close the carbon and safety boundary.** Map severity, quantify immediate emissions and surviving pools, identify life-safety and infrastructure hazards, and record uncertainty.
2. **Protect soil where evidence warrants.** Stabilize priority slopes and drainage paths using treatments supported for the expected rainfall regime. Burned soils can have altered infiltration and runoff response [13]. Mulch can reduce hillslope erosion, but efficacy is treatment- and site-specific [14].
3. **Retain biological legacies when compatible with safety.** Living trees, roots, seed sources, snags, and down wood can support recovery and carbon, but fuel and hazard trade-offs require site decisions. Post-fire logging has reduced natural regeneration and increased short-term fuels in at least one well-studied system [16]; it must not be presumed beneficial or harmful everywhere.
4. **Use contour structures only as tested erosion controls.** Contour-felled logs reduced runoff and sediment for smaller events at several western U.S. sites but did not show treatment effects for larger events, and installation

and degradation mattered [15]. The proposed phrase “pyrogenic contour terrace” is therefore a research hypothesis, not an established carbon intervention.

5. **Treat surplus biomass under a full life-cycle boundary.** Engineered biochar may provide longer-lived storage than unpyrolyzed biomass, but feedstock additionality, process emissions, energy use, product quality, contaminants, counterfactual decay, and double counting must be resolved [10]. Natural charcoal and engineered biochar remain distinct measured pools.
6. **Restore a climate-compatible regenerative pathway.** Use natural regeneration, assisted regeneration, planting, species diversity, and fuel management according to site evidence. Fuel treatments incur near-term carbon costs and their long-run carbon benefit is contingent on fire and drought outcomes [11, 12].
7. **Monitor and adapt.** Re-estimate pool trajectories and model coefficients after each material observation. A policy should be reduced, stopped, or reversed when preregistered ecological or carbon guardrails fail.

This hierarchy operationalizes a minimal-sufficient-intervention principle: intervene only where an identified failure mode, evidence-based action, and monitoring plan justify the ecological cost. The principle is adaptive control, not evidence that less intervention is always better.

7 Implications for carbon markets

Many forest protocols use pooled reserves to compensate for unintentional reversals. The Climate Action Reserve, for example, retires pooled credits for unavoidable reversals such as wildfire and sets project contributions by risk factors [18]. Such designs insure credited tonnes, but a static risk factor can become undercapitalized when climate, severity, portfolio exposure, and uncredited reversal liabilities change [4, 1].

Carbon continuity suggests five design requirements rather than a new credit currency:

1. update disturbance risk dynamically and spatially;
2. monitor all material carbon pools and foregone uptake, not only living biomass at the reversal date;
3. separate measured tonnes from modeled resilience and co-benefits;
4. allow risk-reduction or restoration payments only under additional, verified activity or outcome contracts, without issuing a second claim on the same carbon; and
5. retain ex-post liability, monitoring, and correction pathways after a disturbance rather than terminating observation at the reversal.

A Carbon Continuity Index could inform reserve sizing, covenants, insurance, or stewardship payments. It should not itself be minted as a tonne-denominated offset. Predicted future uptake is not present removal, avoided loss is not the same claim as stored carbon, and biological storage does not erase the distinct durability of fossil carbon released to the atmosphere.

8 Computational and formal verification

The exact program `verification.py` was written before manuscript finalization and its complete output is `verification_output.txt`. It passed 20 of 20 checks. Using exact rational arithmetic, it tested 25,000 parameter cases for the explicit sufficient witness, 25,000 adversarial positive portfolios for necessity, 25,000 boundary constructions, and strict- threshold cases. Negative controls set coupling to zero and searched 10,000 positive portfolios below threshold. Independent closed-form eigenvalue checks agreed with the coupling criterion. The capped trajectory is labeled illustrative and is not part of Theorem 1.

The following Lean claims were frozen before submission and are now proved:

1. threshold sufficiency with the explicit witness $(r, 1 - a)$;
2. threshold necessity from any positive cycle-nondecreasing portfolio;
3. the if-and-only-if coupling theorem;
4. exact stationarity at the boundary;
5. strict durable growth at the explicit witness above the boundary; and
6. a concrete non-vacuity witness at $(a, d, r, p) = (3/5, 4/5, 1/2, 1/5)$.

Aristotle project 40c58299-efeb-4cad-b332-2525b7c8d6d1 completed all six frozen theorem bodies. A separate

local audit rebuilt the returned Lean project, compared every definition and theorem signature with the frozen request, scanned raw and comment-stripped source for proof holes and prohibited constructs, printed the axioms of every theorem, checked the concrete witness, and applied the six-invariant significance gate. All checks passed. The theorems depend only on `propext`, `Classical.choice`, and `Quot.sound`; no `sorry`, `admit`, unsafe declaration, or added axiom remains. This verifies the declared two-pool real-valued model. It does not verify the ecological interpretation, empirical coefficients, treatment effects, market implications, or real-world truth.

9 Limitations

The model has two pools, fixed cycle coefficients, and no explicit fire-arrival process. Aggregation can hide important pool-specific residence times and correlations. Parameters can change with severity, repeated fire, drought, species transition, treatment, and scale. Durable carbon is heterogeneous: standing dead wood, mineral-associated soil carbon, PyC, and harvested products have different hazards and accounting boundaries. The local linear map omits carrying capacity and can imply impossible indefinite growth. CCI depends on a declared capacity or reference and discount factor, so comparisons can be manipulated unless these choices are preregistered. The framework does not model smoke, albedo, methane, nitrous oxide, embodied treatment emissions, or leakage in full. It contains no field dataset, causal identification, cost curve, insurance pricing, or carbon-credit methodology. The novelty search is bounded, not proof of priority.

Most importantly, carbon is only one stewardship objective. A policy that raises modeled CCI may still be unacceptable because of biodiversity, water, air quality, cultural fire practice, public safety, equity, or community rights. Those constraints must remain load-bearing rather than monetized away by a single objective.

10 Conclusion

Forest-carbon permanence should not be framed as the absence of disturbance. It should be tested as the persistence of a mass-balanced carbon service across disturbance-recovery cycles. The Carbon Continuity Framework keeps immediate emissions, post-fire pool transfers, durable carbon, and future uptake in one auditable time model while preserving the distinction between tonnes of carbon and ecological co-benefits.

In the two-pool local model, continuity has an exact threshold: $rp \geq (1 - a)(1 - d)$. This result does not prove that a treatment works. It tells researchers what must be estimated, what can fail, and how living retention, durable retention, transfer, and regenerative support interact. The next scientific step is not a credit product. It is a preregistered longitudinal test against simpler standing-stock baselines using real post-fire carbon-pool and recovery data.

Reproducibility and declarations

Code and evidence availability. The manuscript source, exact verification program and output, generated figure fragments, claim inventory, source ledger, and frozen Lean target package are stored in the Viridis Research control workspace. The source concept PDF is hash-bound in the package but is not scientific evidence. No new field data were collected.

Accountable authorship. Justin D. Hart is the accountable human author and approved the research direction and Aristotle submission request.

Substantive AI-use disclosure. OpenAI Codex extracted and critically reframed the source outline; searched primary and official literature; drafted the manuscript, model, code, figures, claim inventory, and Lean targets; and ran local verification and packaging. Aristotle completed only the frozen Lean proof bodies; Codex then performed the independent local rebuild, statement, safety, axiom, non-vacuity, and significance audits. AI systems are tools and are not authors. Justin D. Hart retains responsibility for claims, interpretation, and any later release.

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validated or offered by this manuscript.

License and status. Internal post-Aristotle working draft. The formal core is independently verified within the stated model. No journal submission, peer acceptance, public DOI, canon admission, empirical validation, treatment validation, market validation, or adoption is claimed.

References

- [1] C. Wu, G. Badgley, M. L. Goulden, et al., “Forest carbon protocols underestimate climate-driven carbon loss risks,” *Nature* 654, 107–113 (2026). doi:10.1038/s41586-026-10571-y.
- [2] W. R. L. Anderegg, A. T. Trugman, G. Badgley, et al., “Climate-driven risks to the climate mitigation potential of forests,” *Science* 368, eaaz7005 (2020). doi:10.1126/science.aaz7005.
- [3] IPCC, “Cross-sectoral perspectives,” in *Climate Change 2022: Mitigation of Climate Change*, Working Group III Contribution to the Sixth Assessment Report, Chapter 12 (Cambridge University Press, 2022). [Official chapter](#).
- [4] G. Badgley, J. Freeman, J. J. Hamman, et al., “California’s forest carbon offsets buffer pool is severely undercapitalized,” *Frontiers in Forests and Global Change* 5, 930426 (2022). doi:10.3389/fgc.2022.930426.
- [5] M. W. Jones, C. Santin, G. R. van der Werf, and S. H. Doerr, “Global fire emissions buffered by the production of pyrogenic carbon,” *Nature Geoscience* 12, 742–747 (2019). doi:10.1038/s41561-019-0403-x.
- [6] S. P. K. Bowring, M. W. Jones, P. Ciais, et al., “Pyrogenic carbon decomposition critical to resolving fire’s role in the Earth system,” *Nature Geoscience* 15, 135–142 (2022). doi:10.1038/s41561-021-00892-0.
- [7] K. S. Hemes, C. A. Norlen, J. A. Wang, M. L. Goulden, and C. B. Field, “The magnitude and pace of photosynthetic recovery after wildfire in California ecosystems,” *Proceedings of the National Academy of Sciences* 120, e2201954120 (2023). doi:10.1073/pnas.2201954120.
- [8] A. F. A. Pellegrini, A. Ahlstrom, S. E. Hobbie, et al., “Fire frequency drives decadal changes in soil carbon and nitrogen and ecosystem productivity,” *Nature* 553, 194–198 (2018). doi:10.1038/nature24668.
- [9] C. Santin, S. H. Doerr, E. S. Kane, et al., “Towards a global assessment of pyrogenic carbon from vegetation fires,” *Global Change Biology* 22, 76–91 (2016). doi:10.1111/gcb.12985.
- [10] J. Lehmann, A. Cowie, C. A. Masiello, et al., “Biochar in climate change mitigation,” *Nature Geoscience* 14, 883–892 (2021). doi:10.1038/s41561-021-00852-8.
- [11] D. E. Foster, J. J. Battles, B. M. Collins, R. A. York, and S. L. Stephens, “Potential wildfire and carbon stability in frequent-fire forests in the Sierra Nevada: trade-offs from a long-term study,” *Ecosphere* 11, e03198 (2020). doi:10.1002/ecs2.3198.
- [12] E. L. Loudermilk, R. M. Scheller, P. J. Weisberg, and A. Kretchun, “Bending the carbon curve: fire management for carbon resilience under climate change,” *Landscape Ecology* 32, 1461–1472 (2017). doi:10.1007/s10980-016-0447-x.
- [13] B. A. Ebel and J. A. Moody, “Synthesis of soil-hydraulic properties and infiltration timescales in wildfire-affected soils,” *Hydrological Processes* 31, 324–340 (2017). doi:10.1002/hyp.10998.
- [14] P. R. Robichaud, S. A. Lewis, J. W. Wagenbrenner, L. E. Ashmun, and R. E. Brown, “Post-fire mulching for runoff and erosion mitigation, Part I: Effectiveness at reducing hillslope erosion rates,” *Catena* 105, 75–92 (2013). doi:10.1016/j.catena.2012.11.015.
- [15] P. R. Robichaud, J. W. Wagenbrenner, R. E. Brown, P. M. Wohlgemuth, and J. L. Beyers, “Evaluating the effectiveness of contour-felled log erosion barriers as a post-fire runoff and erosion mitigation treatment in the western United States,” *International Journal of Wildland Fire* 17, 255–273 (2008). doi:10.1071/WF07032.
- [16] D. C. Donato, J. B. Fontaine, J. L. Campbell, et al., “Post-wildfire logging hinders regeneration and increases fire risk,” *Science* 311, 352 (2006). doi:10.1126/science.1122855.
- [17] U.S. Department of Agriculture, “Quantifying Greenhouse Gas Sources and Sinks in Managed Forest Systems,” in *U.S. Agriculture and Forestry Greenhouse Gas Inventory Methods* (2024), Chapter 5. [Official methods report](#).
- [18] Climate Action Reserve, “Permanence Work Program” (accessed 4 August 2026). [Official program page](#).